INVESTIGATING SLOPE, FAULTS, AND LANDSLIDES RELATIONSHIP USING R: A STUDY ON TAIWAN

Chen, R.[1] & Marsellos, A.E.[2]

[1]North Carolina School of Science and Mathematics, 1219 Broad Street, Durham, NC

[2]Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA

Abstract:

As earthquakes come from the fault activities, soil and loose earth can become dislodged from the surrounding mass of earth. In steeper terrains, the resultant loose earth could fall down, resulting in landslides. As active faults affect how landslide activities occur on steep, mountainous terrains, these two factors should be observed to understand how they affect landslides in mountainous areas. Additionally, factors such as rain also contribute in the creation of loose earth mass as well as in its dislodging from surrounding terrain, further increasing the possibility and frequency of landslides. While anthropological actions, such as tilling farmlands and deforestation, serve to loosen parts of the soil and contribute to landslides, their impacts are hard to quantify, and will thus be excluded from the majority of this study.

Due to Taiwan's mountainous terrain, frequent earthquake activities, and moderate to high level of precipitation, the country is prone to landslides. This research aims to analyze landslide hotspots in Taiwan, primarily through fault locations and slope steepness. Data from GEM Global Faults Database was used in plotting the active faults of Taiwan while elevatr provided for a way to calculate elevations and slope angles of Taiwan's mountains. It was found that the western border of Taiwan's mountainous region were especially vulnerable to landslides, characterized by both slope difference and many active faults. Future studies could develop the roles of human action into quantifiable variables which may be expressed by models.

Background Information:

- Taiwan has experienced multiple large-scale landslides, with both severity and frequency worsening
- Due to climate change, there has been an increase in the severity and frequency of storms, increasing rainfall and fault activities
- Active faults can cause landslides directly (loosening earth) and indirectly (cycles of damming rivers and feedback to fault activity) (Fig 1)
- Slope and rain activity can compound on landslide activity, but usually happens more gradually

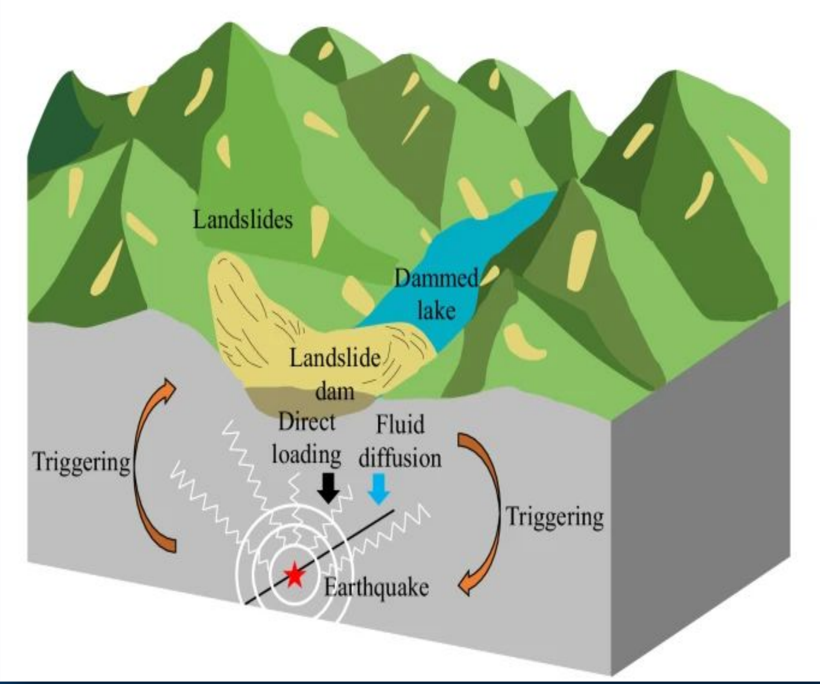


Figure 1: The bidirectional causation of earthquakes (from faults) and dammed lakes (from landslides)

Method/Materials:

- **RStudio (v.4.2.1)**
  - Packages used: sf, terra, elevatr, rnaturalearth, viridis, ggplot2
  - Used data from GEM Faults Database for plotting fault lines
  - Plotted the shape of Taiwan and its elevation using rnaturalearth
  - Calculated slope angles from the elevations
- Taiwan Precipitation Map
  - Used in conjunction with fault and elevation to evaluate landslide susceptibility
- Custom overlaying was done to combine the fault data with Taiwan's DEM in order to resolve problems with built-in overlaying methods

Results:

- Slope calculation used to generate slope map from elevation
- Individual slope, elevation, and active fault maps for coordinate system around Taiwan (Fig 2,3,4)
- Overlaying of GEM Database fault maps with generated slope maps from different coordinate systems
- Separation of fault types on overlay map

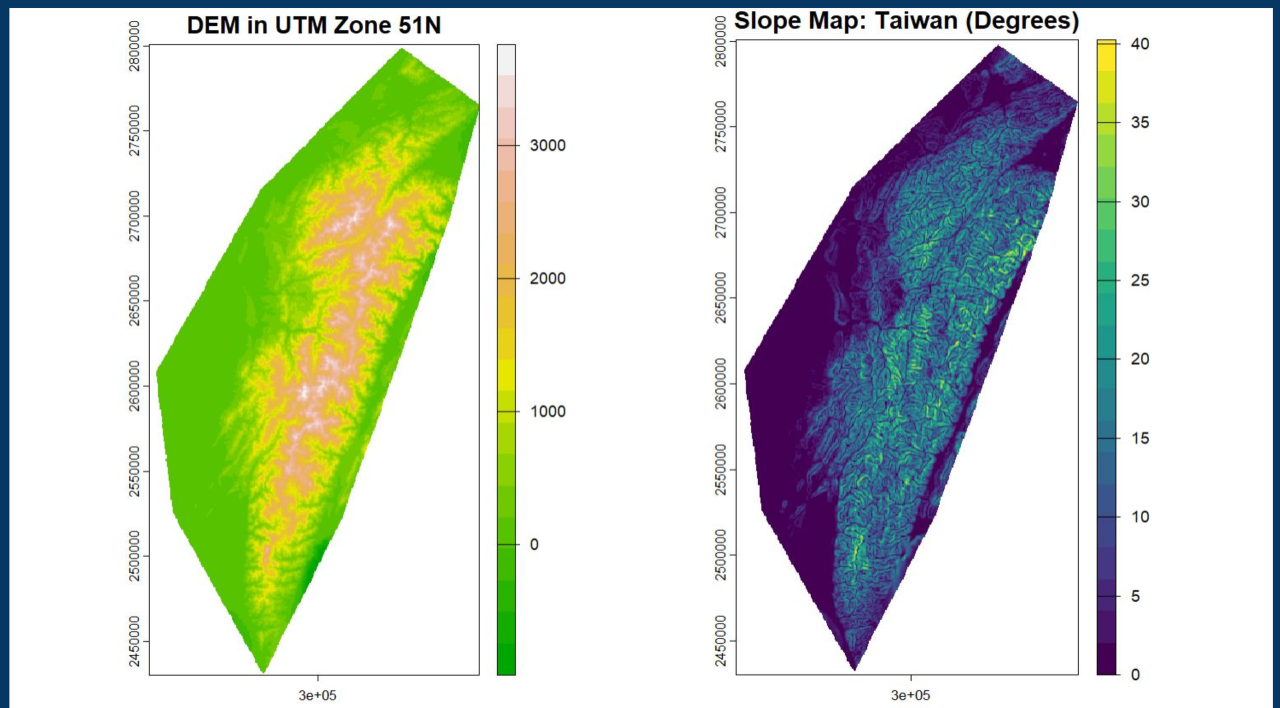
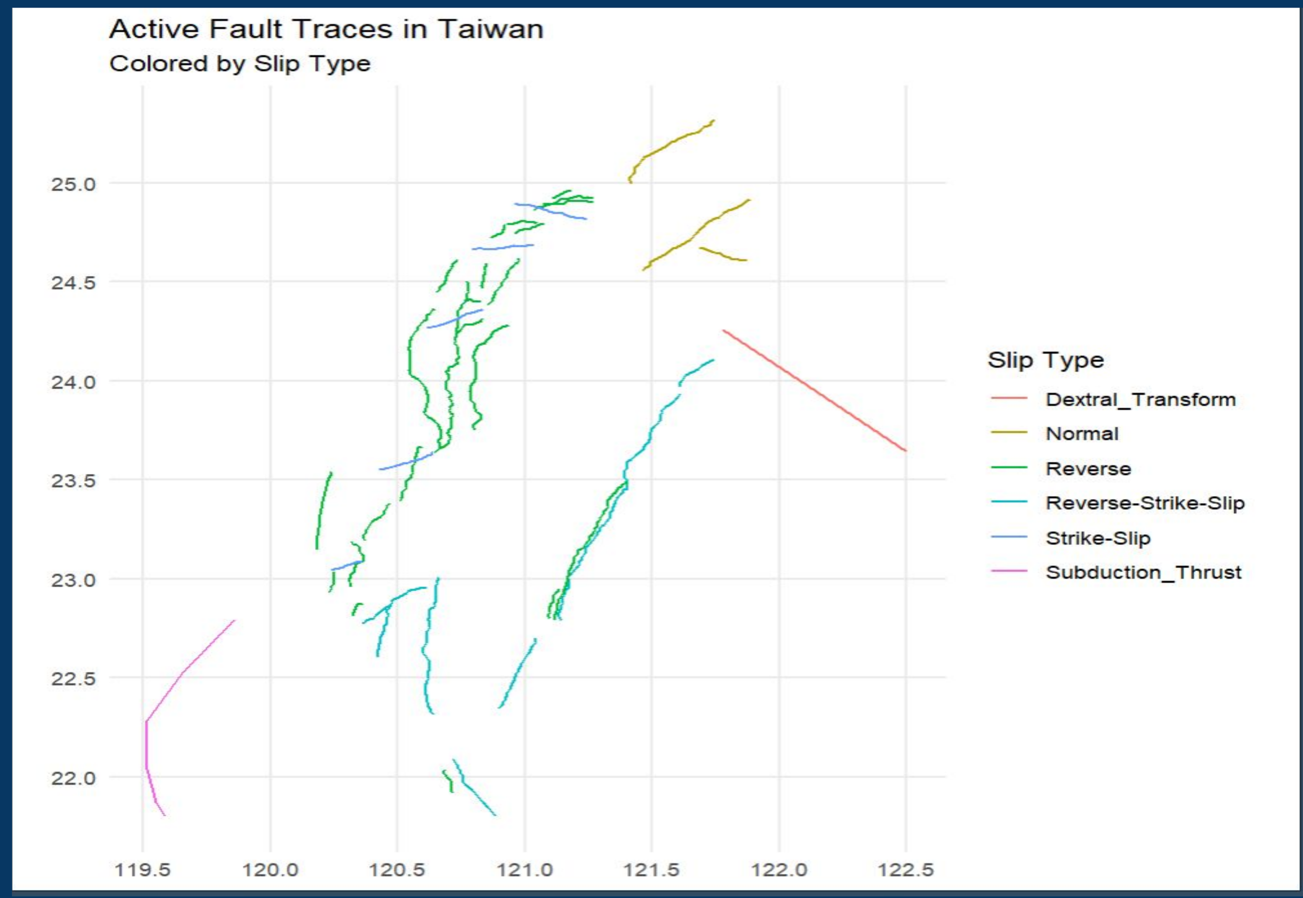


Figure 2&3: Elevation and Slope maps of Taiwan



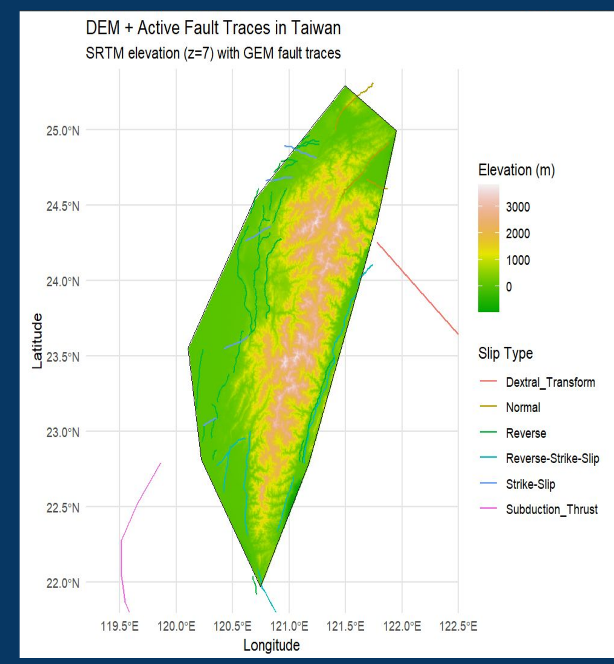


Figure 4: Map of Taiwan's active trace faults shown based on slip types;

Figure 5: Digital Elevation Model (DEM) of Taiwan's slope and elevation distributions with active faults types overlayed

Can be seen that Southeast Taiwan, as well as much of its western area bordering the mountains are the most susceptible to landslides, with the east border to the mountains also having substantial risk.

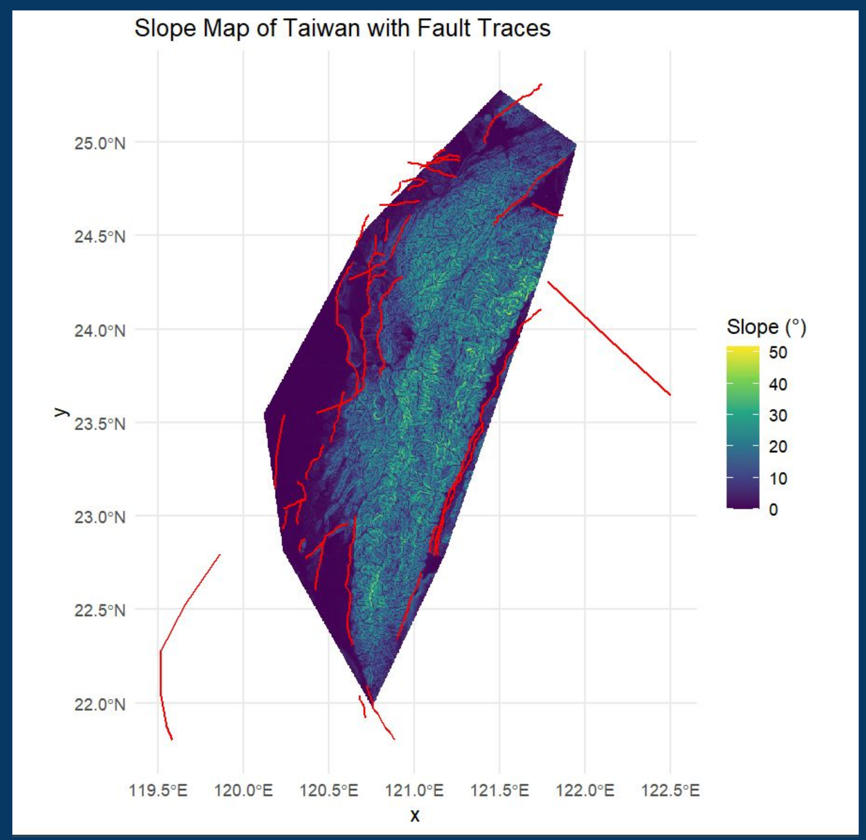


Figure 6: Visualization of Taiwan's fault lines and slope overlay; map plotting rain patterns across Taiwan

Discussion/Conclusion:

- Taiwan overlaps most with reverse and strike-slip faults (Fig 6)
  - Strike slip faults generate the seismic activity that directly causes landslides
  - Reverse faults create steep terrains that increase risk of landslides
- GIS: high elevation —> increased likelihood of landslides (Fig 2&3)
  - Eastern border of Taiwan and Central-western Taiwan are most affected
- Cities such as Taitung, Hualien, and Taichung are at the edge of steeper-sloped terrain and line up with faults (Fig 5)
- Data can help creation of active landslide warning services

Acknowledgements:

I would like to thank Dr. Antonios E. Marsellos for providing me with the necessary tools and guidance to conduct this research.

Credit Authorship Contribution Statement:

Chen, R: Rstudio coding, All figures, poster writing and editing; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing;

References:

Lee, Chyi-Tyi. "Statistical Seismic Landslide Hazard Analysis: An Example from Taiwan." *Engineering Geology*, vol. 182, Nov. 2014, pp. 201–212, https://doi.org/10.1016/j.enggeo.2014.07.023.

Chen, Chi-Wen, et al. "Relationship between Landslide Size and Rainfall Conditions in Taiwan." *Landslides*, vol. 14, no. 3, 26 Dec. 2016, pp. 1235–1240, link.springer.com/article/10.1007%2Fs10346-016-0790-7, https://doi.org/10.1007/s10346-016-0790-7.

Lin, S. C., et al. "Evolution of Landslide Hotspots in Taiwan." *Landslides*, vol. 14, no. 4, 9 Apr. 2017, pp. 1491–1501, https://doi.org/10.1007/s10346-017-0816-9.

Tsai, Yuan-Jung, et al. "Rainfall-Induced Large-Scale Landslide Hazard Zonation in Taiwan." *Workshop on World Landslide Forum*, 11 June 2017, pp. 719–732, www.researchgate.net/publication/318152660_Rainfall-Induced_Large-Scale_Landslide_Hazard_Zonation_in_Taiwan, https://doi.org/10.1007/978-3-319-53498-5_83.

Zhang, Zhen, et al. "Landslide Hazard Cascades Can Trigger Earthquakes." *Nature Communications*, vol. 15, no. 1, 8 Apr. 2024, https://doi.org/10.1038/s41467-024-47130-w.

Styron, Richard, and Marco Pagani. "The GEM Global Active Faults Database." Earthquake Spectra, vol. 36, no. 1_suppl, Oct. 2020, pp. 160–180, doi:10.1177/8755293020944182

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., ... & Yutani, H. (2019). Welcome to the Tidyverse. Journal of Open Source Software, 4(43), 1686.

Pebesma, E. (2018). Simple Features for R: Standardized Support for Spatial Vector Data. The R Journal, 10(1), 439-446.

Massicotte P, South A (2025). rnaturalearth: World Map Data from Natural Earth. R package version 1.1.0.9000, https://github.com/ropensci/rnaturalearth.

Hollister J, Shah T, Nowosad J, Robitaille A, Beck M, Johnson M (2025). elevatr: Access Elevation Data from Various APIs. doi:10.5281/zenodo.8335450, R package version 0.99.1, https://github.com/usepa/elevatr/.

Hijmans R (2025). *terra: Spatial Data Analysis*. R package version 1.8-72, https://github.com/rspatial/terra.